

A 01 Sequence

退出全屏

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Given a binary cyclic sequence S of length n , whose elements are either 0 or 1, you can do the following operation any number of times.

- Operation:** if the length of S is greater than or equal to 3, choose a position i ($1 \leq i \leq n$) such that $S_i = 1$, remove the element on position i and both its adjacent elements (remove 3 elements in total), and do not change the relative order of the other elements. Note that elements in the first and last positions are considered adjacent.

A binary cyclic sequence S is called *good* if you can make S empty using the operation above.

And the *beauty* of a binary cyclic sequence S , $f(S)$, is defined as the minimum number of modifications needed to make S good. In a single modification you can flip an arbitrary element in S , that is, 0 becomes 1 and 1 becomes 0.

Given are a binary string a of length n and q queries. For the i -th query you are given two integers l_i and r_i , and you should answer $f(a_{l_i..r_i})$ (where we consider the substring $a_{l_i..r_i}$ as a cyclic sequence).

输入格式:

The first line contains two integers n and q ($3 \leq n \leq 10^6, 1 \leq q \leq 10^6$) — the length of string a and the number of queries, respectively.

The second line contains the string $a_1a_2 \cdots a_n$, where each a_i is either 0 or 1.

Each of the following q lines contains two integers l_i and r_i ($1 \leq l_i \leq r_i \leq n, (r_i - l_i + 1) \equiv 0 \pmod 3$) describing the i -th query.

输出格式

Output q lines, where the i -th line contains a single integer — the answer for the i -th query.

输入样例:

```
7 7
1000011
1 3
2 4
3 5
4 6
5 7
1 6
2 7
```

```
12 1
110110110000
1 12
```

```
20 10
00001000100000010000
18 20
14 16
7 12
2 10
16 18
6 20
8 10
13 15
1 6
1 12
```



输出样例:

```
0
1
1
0
0
1
1
```

```
1
```

```
1
0
1
1
0
3
0
1
1
2
```

代码长度限制	16 KB
时间限制	2000 ms
内存限制	512 MB
栈限制	128000 KB

B Power Station

[退出全屏](#)

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You want to build n power stations.

The i -th power station requires 1 unit of material to build it up, and the process of building a power station takes no time. Assume that the i -th power station is built up in time t , then the power station will produce b_i units of material at time $t + a_i, t + 2a_i, t + 3a_i, \dots$ and so on.

Initially, you have 1 unit of material, and you want to know how much time it takes at least to build all n power stations.

输入格式:

The first line contains an integer n .

Each of the following n lines contains two integers a_i, b_i , representing the information of the i -th power station.

It is guaranteed that

$$1 \leq n \leq 16, 1 \leq a_i \leq 10^6, 1 \leq b_i \leq n.$$

输出格式

Output a single integer, representing the minimum time it takes to build all n power stations.

输入样例:

```
4
3 3
2 1
2 1
2 1
```

```
5
181077 1
213330 1
290524 2
265870 2
309796 2
```

输出样例:

3

446947

代码长度限制	16 KB
时间限制	3000 ms
内存限制	256 MB
栈限制	131072 KB

C Delete the Tree

[退出全屏](#)

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Alice wants to delete a tree.

She can do the following operations:

- **Delete**: Delete a vertex in the tree and all the edges connected to it.
- **Shrink**: Choose a vertex x that has exactly two vertices connected to it; suppose the vertices are u, v , Alice can delete x and the edges $(u, x), (x, v)$ and add a new edge (u, v) .

Alice wants to achieve her goal using the least number of **Delete** operations. Please help her.

输入格式:

The first line contains one integer T ($1 \leq T \leq 10^6$), denoting the number of test cases.

For each test case:

The first line contains one integer n ($1 \leq n \leq 10^6$), denoting the size of the tree.

Then $n - 1$ lines follow, each containing two integers u, v , denoting the edge that connects u and v on the tree.

It is guaranteed that the sum of n will not exceed 10^6 .

输出格式

For each test cases, output a single integer, denoting the least number of **Delete** operations that Alice must use.

输入样例:

```
2
2
1 2
5
1 2
1 3
2 4
2 5
```

输出样例:

```
2
3
```

代码长度限制

16 KB

时间限制

1000 ms

内存限制

512 MB

栈限制

131072 KB

D Find the Number

[退出全屏](#)

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Let's call a number x good if x is an interger and $\text{ctz}(x) = \text{popcount}(x)$.

$\text{ctz}(x)$ is the number of trailing zeros in the binary representation of x .

$\text{popcount}(x)$ is the number of 1's in the binary representation of x .

For example:

$\text{ctz}(12) = \text{ctz}(1100_2) = 2$ and $\text{popcount}(12) = \text{popcount}(1100_2) = 2$, so 12 is a good number.

Now you are given an interval $[l, r]$, you need to find a good number in it. If there are multiple solutions, print any of them; if there is no solution, print -1 instead.

输入格式:

The first line contains a single integer T ($1 \leq T \leq 10^5$) --- the number of test cases.

For each test case, there is one single line containing two integers l, r ($1 \leq l \leq r \leq 10^9$) --- the interval.

输出格式

For each test case, print one line containing the answer.

If there are multiple solutions, print any of them; if there is no solution, print -1 instead.

输入样例:

```
5
38 47
57 86
23 24
72 83
32 33
```

输出样例:

```
-1
68
-1
-1
-1
```

代码长度限制	16 KB
时间限制	1000 ms
内存限制	512 MB
栈限制	131072 KB

E Tower Defense

[退出全屏](#)

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Rikka is playing a 2D tower defense game. In each game, many towers and enemies are generated on the map. Each of them can be regarded as a point in a 2D Cartesian coordinate system. Rikka's task is to eliminate enemies using the towers.

At the beginning of the game, n towers are generated. In each round, an enemy occurs on the map. The enemy never moves, and Rikka can eliminate it using the following two actions **any times**.

- **Build a tower.** Rikka can select any two existing towers and build a new tower on the line segment between them. Formally, suppose the coordinates of the two selected towers are (x_1, y_1) and (x_2, y_2) . Rikka can select a **real number** $\alpha \in [0, 1]$ and then build a tower on point $(\alpha x_1 + (1 - \alpha)x_2, \alpha y_1 + (1 - \alpha)y_2)$.
- **Eliminate the enemy.** If the enemy is on the line segment between two towers, Rikka can attack and eliminate it. Formally, suppose the coordinate of the enemy is (x, y) . Rikka can eliminate the enemy if (1) there exist two towers with coordinates (x_1, y_1) , (x_2, y_2) , and (2) there exists a **real number** $\alpha \in [0, 1]$ such that $x = \alpha x_1 + (1 - \alpha)x_2, y = \alpha y_1 + (1 - \alpha)y_2$.

Rikka can skip the round at any time. A round will end immediately after the enemy is eliminated or Rikka chooses to skip. After that, **both the current enemy and all towers built in this round will disappear from the map**, and then a new round will start with **only the initial n towers remaining**.

In each round, if the enemy is successfully eliminated, the cost will be the number of actions taken by Rikka. Otherwise, the cost will be 10^9 .

Now, Rikka has been a master of this game. She can always take the optimal action that minimizes the cost. To have more fun, Rikka plans to increase the difficulty by banning some initial towers. Clearly, there are 2^n different possibilities of banning towers. To measure the increased difficulty, Rikka shows you a game and wants you to simulate all these possibilities: In each round, Rikka wants you to calculate the cost she will use for each possibility and tell her the total cost.

输入格式：

The first line contains two integers n, m ($1 \leq n, m \leq 1000$), representing the number of initial towers and rounds.

Then n lines follow. Each line contains two integers x, y ($-10^9 \leq x, y \leq 10^9$), representing the coordinate of each initial tower.

Then m lines follow. Each line contains two integers x, y ($-10^9 \leq x, y \leq 10^9$), representing the coordinate of the enemy that occurs in each round.

The input guarantees that the given $n + m$ points are pairwise different.

输出格式

For each round, output a single number, the sum of costs for all possibilities of banning initial towers.

The answer may be very large. You need only to output the answer module $10^9 + 7$.

输入样例：

```
3 2
0 0
2 0
1 2
1 1
1 0
```

```
4 1
0 0
2 0
2 2
0 2
1 1
```

输出样例：

```
999999960
999999967
```

```
999999951
```

样例解释

In the first round of the first sample, the cost will be 2 if no tower is banned. In the other 7 cases, the cost will be 10^9 . Therefore, the answer is 7000000002.

In the second round of the first sample, the cost will be 1 if no tower is banned or only the tower at $(1, 2)$ is banned. In the other 6 cases, the cost will be 10^9 . Therefore, the answer is 6000000002.

代码长度限制	16 KB
时间限制	1000 ms
内存限制	256 MB
栈限制	131072 KB

F Bacteria

退出全屏

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Doctor P is currently investigating the growth of a special type of bacteria.

Initially, on Day 1, there are some bacteria in the Petri dish.

Every day, every bacterium reproduces the same number of offsprings (possibly zero).

Doctor P finds that on Day k , the number of bacteria does not exceed n . He is wondering the number of distinct bacteria number sequences over the k days.

Formally speaking, you need to count the total number of positive integer sequences, where each sequence $a_1, a_2, \dots, a_k$ satisfies that $a_i \mid a_{i+1}$ for $1 \leq i < k$ and $a_k \leq n$.

输入格式:

The only line of input will contain two space-separated integers n and k ($1 \leq n, k \leq 10^9$).

输出格式

Output a single integer representing the number of distinct bacteria number sequences modulo $10^9 + 7$.

输入样例:

```
4 2
6 5
```

输出样例:

```
8
56
```

代码长度限制	16 KB
时间限制	1000 ms
内存限制	128 MB
栈限制	131072 KB

G Read the Documentation

退出全屏

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Yunheec is competing in Coffee Cup Pouring Competition (CCPC).

The competition lasts n seconds. In each second, Yunheec is asked to pour some kind of liquid out from the coffee cup. Yunheec does not know what kind of liquid he is supposed to pour, so he needs to read the documentation. Yunheec is able to choose whether he is going to read the documentation during each second.

However, Yunheec is competing under close observation, and the observer does not like Yunheec reading the documentation. Particularly, if Yunheec has been reading the documentation for consecutive 5 seconds, he will be disqualified.

Given are integers $T, x_1, \dots, x_4, a_1, \dots, a_n$.

Let the anger of the observer be X , and the score of Yunheec be S , initially $X = S = 0$. For each i ($1 \leq i \leq n$), if Yunheec chose to read the documentation during second i , the following would happen at the end of second i :

- Let t be the number of seconds Yunheec has been continuously reading the documentation for. For example, $i = 5$ and Yunheec read the documentation during seconds 3, 4, 5, but not second 2, then $t = 3$.
- If t is less than or equal to 4, x_t is added to X .
- Then, if t is greater than or equal to 5, or X is strictly greater than T , the observer gets angry, S is set to zero and the competition immediately ends. Otherwise, a_i is added to S .

The final value of S will be the score of Yunheec.

Find the maximum score Yunheec can get.

输入格式:

The first line contains two integers n and T ($1 \leq n \leq 100, 1 \leq T \leq 10^9$) — the duration of the competition and the anger needed for the observer to get angry.

The second line contains four integers x_1, x_2, x_3, x_4 ($1 \leq x_i \leq 10^9$).

The third line contains n integers $a_1, \dots, a_n$ ($1 \leq a_i \leq 10^9$).

输出格式

Output the maximum score Yunheec can get.

输入样例:

```
6 11
1 2 4 8
1 10 100 1000 10000 100000
```

1 1
67 67 88 67
20220911

10 808995774
227150190 224559951 139457454 87915609
96204862 799843967 446173607 796619138 402690754 223219513 668171337 312183499 905549873 673542337

输出样例:

111011

0

2559447046

代码长度限制	16 KB
时间限制	2500 ms
内存限制	512 MB
栈限制	128000 KB

H Step Debugging

退出全屏

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Rikka is a contestant in C-- Collegiate Programming Contest (CCPC), a competition for writing programs using a popular language C--.

A C-- expression is inductively defined in the following way.

- `arithmetic` performs some arithmetic calculations.
- `library` invokes a function in the standard library of C--.
- For any C-- expression e and integer $w \in [1, 100]$, `repeat  $e$  for  $w$  times` repeats expression e for w times.
- For any two C-- expressions e_1, e_2 , $e_1\ e_2$ runs expressions e_1, e_2 in order.

A C-- program is a C-- expression followed by token `fin`, representing the end of the program.

Today, Rikka takes part in the online selection of CCPC. She writes a C-- program, but it fails in passing all test cases. Now, Rikka wants to figure out what goes wrong in this program using step debugging. However, such a choice is risky: Stepping into a library function of C-- may be judged as cheating.

Therefore, before debugging, Rikka wants to mature the risk of stepping into library functions. She wants to calculate how many times the C-- program invokes library functions.

输入格式:

The input contains a single line, representing the C-- program written by Rikka.

The input guarantees the following four points.

- The input C-- program is valid.
- The input file is no larger than 10kB.
- There is no extra space. Concretely, there is no leading space and succeeding space, and there is exactly one space between any two tokens.
- There is a line break right after the end of the program, i.e., token `fin`.

输出格式

Output a single number, representing the times of invoking library functions.

The answer may be very large. You only need to output the answer module 20220911.

输入样例:

library arithmetic library library fin

repeat library arithmetic for 5 times library arithmetic fin

repeat library repeat repeat library for 3 times arithmetic library for 3 times for 100 times fin

repeat library repeat arithmetic library for 3 times repeat library for 2 times arithmetic for 4 times arithmetic fin

输出样例:

3

6

1300

24

代码长度限制	16 KB
时间限制	1000 ms
内存限制	256 MB
栈限制	131072 KB

I Permutation

退出全屏

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We can sort all permutations of length n in lexicographical order and thus give each permutation a rank. For example, when $n = 5$, the rank of 1 2 3 5 4 is 2, and the rank of 5 4 3 2 1 is 120.

Alice has a permutation $p_1, p_2, \dots, p_n$. She can also shift a permutation; that is to say, she can transfer a permutation $q_1, q_2, q_3, \dots, q_m$ to $q_2, q_3, \dots, q_m, q_1$. Using this, she can construct the following n different permutations:

- $p_1, p_2, p_3, \dots, p_{n-2}, p_{n-1}, p_n$
- $p_2, p_3, p_4, \dots, p_{n-1}, p_n, p_1$
- $\dots$
- $p_n, p_1, p_2, \dots, p_{n-3}, p_{n-2}, p_{n-1}$

Now Alice wants to know the sum of the ranks of these permutations.
Since the answer may be too large, she only need you to tell her the result modulo 998244353.

输入格式:

The first line contains one integer n ($1 \leq n \leq 2.5 \times 10^5$), denoting the length of the permutation.
The next line contains n integers $p_1, p_2, \dots, p_n$, denoting the permutation.

输出格式

A single integer, denoting the sum of the ranks modulo 998244353.

输入样例:

```
5
1 2 3 5 4
```

输出样例:

```
297
```

代码长度限制	16 KB
时间限制	7500 ms
内存限制	512 MB
栈限制	131072 KB

J Gachapon

退出全屏

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You are designing a card-drawing system. And the best card you can draw out is called SSR.

Given X and Y , You want to design a card-drawing system that satisfies the following conditions:

- Assume that after drawing the last SSR, you have not drawn out any SSR for i consecutive draws, then the probability that you will draw out an SSR in the next draw (the $(i + 1)$ -th draw after the last SSR) is a_{i+1} .
- At least one SSR will be drawn out in any X consecutive draws, which means that $a_X = 1$.
- It is possible that there may not have any SSR in $X - 1$ consecutive draws, which means that $\forall 1 \leq i < X, a_i < 1$.
- For any possible draw, the probability that an SSR is drawn out is not zero, which means that $\forall 1 \leq i, 0 < a_i$.
- Assume that y is a random variable representing the number of draws you take to draw out an SSR, then $E[y] = Y$.

Find any fraction array a_i that satisfies the conditions above. Also, the absolute value of the numerator and denominator should not be too large.

输入格式:

The first line contains two integers X, Y . It is guaranteed that $2 \leq Y < X \leq 100$.

输出格式

Output X lines. The i -th line contains two integers x, y , and the value of a_i is exactly x/y . You should guarantee that $\gcd(x, y) = 1$ and $0 \leq x, y \leq 10\,000$.

It is guaranteed that a solution always exists under the constraints above.

输入样例：

```
3 2
6 2
```

输出样例：

```
1 3
1 2
1 1

2 3
1 5
1 4
1 3
1 2
1 1
```

代码长度限制	16 KB
时间限制	1000 ms
内存限制	256 MB
栈限制	131072 KB

K Pyramad

退出全屏

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Pyramad is a hero in game *Battlegrounds*. Recently, it has been re-designed, and its mechanism has become much more complex. Therefore, Rikka wants to get familiar with the new version via some simple practice.

In the practice, Rikka controls a minion with an initial attribute equal to 1. The practice proceeds for n rounds. In each round, Rikka first improves the attribute of the minion using the skill of *Pyramad*, and then a battle will break out. Rikka can win the battle in the i -th round if and only if the attribute of the minion is at least x_i .

The skill of *Pyramad* is parametrized by an integer w , whose initial value is 1. Each time when the skill is used, Rikka can choose one from the following two actions.

- Increase the value of w by 1 **permanently**.
- Increase the attribute of the minion by w .

Rikka wants you to determine the optimal strategy and tell her the maximum number of battles she can win.

输入格式：

The first line contains a single integer $n(1 \leq n \leq 5000)$, representing the number of rounds in the practice.

The second line contains n integers $x_i(1 \leq x_i \leq 5000)$, representing the smallest attribute for winning the i -th battle.

输出格式

Output a single integer, the maximum number of battles Rikka can win if she plays optimally.

输入样例：

```
5
2 3 4 6 8
```

```
20
1 2 4 6 4 19 9 3 31 5 26 16 4 53 76 53 19 101 2 108
```

输出样例：

```
4
13
```

样例解释

- The following shows an optimal strategy for the first sample.
- In the first round, Rikka chooses action 2, the attribute of the minion becomes 2, and Rikka wins the battle.
 - In the second round, Rikka chooses action 1, the attribute of the minion remains unchanged, and Rikka loses the battle.
 - In the following three rounds, Rikka always chooses action 2. At this time, the attribute of the minion becomes 4, 6, 8 respectively, and Rikka wins all three battles.

代码长度限制	16 KB
时间限制	1000 ms
内存限制	256 MB
栈限制	131072 KB

L LCS-like Problem

退出全屏

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Given string s , we use $|s|$ to represent the length of the string.

Given string s , we call string t a subsequence of s iff:

- There exists $1 \leq a_1 < a_2 < \cdots < a_{|t|} \leq |s|$, so that for every $1 \leq i \leq |t|$, $s_{a_i} = t_i$.

Given strings s and t , we call string p a common subsequence of s and t iff:

- p is a subsequence of both s and t .

The problem you have to solve is that: Given strings s and t , find the longest string s' that satisfies the following properties:

- s' is a subsequence of s .
- For any string p being a common subsequence of s' and t , it always holds that $|p| \leq 1$.

Because s' may not be unique, you just need to output the length of s' .

输入格式:

The first line has a string s , which contains only lowercase letters a to z . It is guaranteed that $1 \leq |s| \leq 5 \times 10^5$.

The second line has a string t , which contains only lowercase letters a to z . It is guaranteed that $1 \leq |t| \leq 5 \times 10^5$.

输出格式

Output a single number, which represents the length of s' .

输入样例:

```
apple
grape

sakana
chinanago
```

输出样例:

```
3

3
```

样例解释

In sample 1, we can choose $s' = \text{ppl}$

In sample 2, we can choose $s' = \text{sak}$ or $s' = \text{skn}$

代码长度限制	16 KB
时间限制	1000 ms
内存限制	256 MB
栈限制	131072 KB